Some Basic Concepts of Chemistry

SHORT NOTES

Important Formulae of Concentration Terms

1. $w/w\ \% = \dfrac{w_2}{w_1 + w_2} \times 100$

2. $w/v\ \% = \dfrac{w_2}{V_{sol}} \times 100$

3. $v/v\ \% = \dfrac{v_2}{v_1 + v_2} \times 100$

4. $ppm = \dfrac{w_2}{w_1 + w_2} \times 10^6$

5. $ppb = \dfrac{w_2}{w_1 + w_2} \times 10^9$

6. $X_2 = \dfrac{n_2}{n_1 + n_2}; X_1 = \dfrac{n_1}{n_1 + n_2}$

7. $M = \dfrac{W_2 \times 1000}{M_2 \times V(ml)}$

8. $M = \dfrac{\frac{w}{w}\% \times d \times 10}{\text{Molar mass of solute}}$

9. $m = \dfrac{\text{Moles of the solute}}{\text{mass of solvent(kg)}} = \dfrac{W_2 \times 1000}{M_2 \times W_1(g)}$

10. $m = \dfrac{x_2 \times 1000}{x_1 \times Mw_1(g)}$

11. $d = M\left(\dfrac{1}{m} + \dfrac{M_2(g)}{1000}\right)$

12. $M = M\dfrac{1000dX_2}{x_1M_1 + x_2M_2(g)}$

Conc. Terms	Formula
Mole Fraction	Mole fraction $= \dfrac{\text{Moles of component}}{\text{Total number of moles present in solution}}$ $x_A = \dfrac{n_A}{n_A + n_B + n_C .....}$
Parts per million	ppm of a solute in solution $= \dfrac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$
Degree of hardness of water	Hardness of water $= \dfrac{\text{mass of } CaCO_3}{\text{Total mass of water}} \times 10^6$
Mass Percentage	$\%w/w = \dfrac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$
Mass by volume Percentage	$\%w/v = \dfrac{\text{Mass of solute}}{\text{Volume of solution}} \times 100$
Volume by volume percentage	$\%v/v = \dfrac{\text{Volume of solute}(cm^3)}{\text{Volume of solution}(cm^3)} \times 100$
Molarity	Molarity, (M) $= \dfrac{\text{Moles of solute}}{\text{Volume of solution (L)}}$
Molality	Molality, (m) $= \dfrac{\text{Moles of solute}}{\text{Mass of solvent (Kg)}}$